Independent Audit of the One-Body Dispersion Test and Systematic Hunt for the Factor of Four

Independent Audit Note

November 4, 2026

Abstract

We conduct an independent mathematical and numerical audit of `phase2/onebody_dispersion_test.tex` and its supporting scripts (`phase2/scripts/onebody_continuum.py` and `phase2/scripts/onebody_toda.py`), which compare the small-amplitude $O(\hbar^0)$ one-body soliton dispersion correction at fixed momentum between the formally continued continuum projected determinant ($C_{\rm ac} = +2/\pi \approx 0.63662$) and the quantum Toda Fermi-edge limit (0.15912 ± 0.00334). First, we re-derive the continuum bad-equation continuation from scratch, confirming both sign bugs exposed during the test: `phase2/report.tex` prints the bad-equation Lagrangian with the wrong sign on the cubic term ($+\phi_x^3$ instead of the required $-\phi_x^3$), and `phase2/oneloop.tex` Eq. (28) carries the wrong sign on the bad-equation Wick normal-ordering tadpole ($+3\hbar_B k/\pi$ instead of $-3\hbar_B k/\pi$). Second, we prove four exact non-perturbative identities for the finite-sea Toda Bethe equations—including $2at\,\rho(q)\eta(q) = 1$ and $v_s(a,h) = \frac{a}{2}(q - h\,\partial_h q|_a)$—which determine the Toda Fermi-edge limit analytically in closed form: the quantum sound-speed shift is $s_{1,T} = \frac{3}{4\pi}$ and the one-body dispersion slope is $C_T = \frac{3}{4\pi} - \frac{1}{4\pi} = \frac{1}{2\pi} = 0.15915494309\ldots$, in both the fixed-$y$ and fixed-P orders of limits. The numerical extraction in `onebody_toda.py` is non-circular and its error bars honestly enclose $\frac{1}{2\pi}$. Third, we audit all six normalisation candidates in the Toda–Boussinesq dictionary; each introduces a conversion factor of 1. We therefore conclude **(b)**: no missing factor of 4 exists in the dictionary, and the exact ratio $C_{\rm ac}/C_T = (2/\pi)/(1/(2\pi)) = 4$ (or -8 after subtracting the Toda sound-speed shift) is a genuine structural discrepancy between continuum short-wave dispersive modes ($|q| \sim \sqrt{3}/2$) and the Toda lattice Fermi sea.

1 Scope, conventions, and verification script

Claim labels follow `AGENTS.md`: **(a)** proved in the literature; **(b)** conjectured or heuristic in the literature; **(c)** derived and verified in this project; **(d)** a guess or formal continuation assumption.

Unless stated otherwise, the baseline convention is the *good* Boussinesq equation

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \phi_x = 2(\log\tau)_{xx}, \qquad \mathcal{L}_g = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \phi_x^3 - \tfrac{1}{2}\phi_{xx}^2, \tag{1}$$

with canonical momentum $\pi = \phi_t$ and symplectic two-form $\Omega_B = \int dx\, d\pi \wedge d\phi$. The *bad* Boussinesq equation is

$$u_{tt} = u_{xx} + u_{xxxx} + 3(u^2)_{xx}, \qquad u = \phi_x, \tag{2}$$

with one-soliton data $k = \sinh\beta$, $v = \varepsilon\cosh\beta$, $u_0(x,t) = \frac{k^2}{2}\operatorname{sech}^2\frac{k(x-vt)}{2}$, mass $M = \int u_0\,dx = 2k$, field momentum $P_0 = -\int \pi_0\phi_{0,x}\,dx = \frac{2}{3}\varepsilon k^3\cosh\beta = \frac{2}{3}\varepsilon\sinh^3\beta\cosh\beta$, and energy $E_0 = \frac{2}{3}\sinh^3\beta + \frac{8}{15}\sinh^5\beta$.

All symbolic and 65-digit numerical checks reported in this audit are executed in `phase2/scripts/audit_oneb` which verifies SHA-256 hashes of all audited repository files to guarantee zero modifications and ends by printing `ALL CHECKS PASSED`.

2 Audit of the continuum bad-equation dispersion correction

2.1 Bad-equation Lagrangian sign error in `phase2/report.tex` and tadpole sign error in `phase2/oneloop.tex`

1. Cubic sign error in `phase2/report.tex` (c). For a potential Lagrangian density $\mathcal{L}(\phi_t, \phi_x, \phi_{xx}) = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 + \lambda\phi_x^3 + \frac{1}{2}\phi_{xx}^2$, the Euler–Lagrange equation is

$$0 = -\partial_t \frac{\partial \mathcal{L}}{\partial \phi_t} - \partial_x \frac{\partial \mathcal{L}}{\partial \phi_x} + \partial_x^2 \frac{\partial \mathcal{L}}{\partial \phi_{xx}} = -\phi_{tt} + \phi_{xx} - 3\lambda(\phi_x^2)_x + \phi_{xxxx}. \tag{3}$$

Differentiating with respect to x and setting $u = \phi_x$ yields $u_{tt} = u_{xx} + u_{xxxx} - 3\lambda(u^2)_{xx}$ **(c)** (`audit_onebody.py`). Matching the bad equation (2) uniquely requires $\lambda = -1$, so the correct bad-equation Lagrangian and Hamiltonian densities are **(c)**

$$\mathcal{L}_b = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 - \phi_x^3 + \tfrac{1}{2}\phi_{xx}^2, \qquad \mathcal{H}_b = \pi\phi_t - \mathcal{L}_b = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}u^2 + u^3 - \tfrac{1}{2}u_x^2. \tag{4}$$

By contrast, `phase2/report.tex` (line 69) prints $\mathcal{L}_{\text{report}} = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 + \phi_x^3 + \frac{1}{2}\phi_{xx}^2$, which generates $u_{tt} = u_{xx} + u_{xxxx} - 3(u^2)_{xx}$ and fails the bad equation on the one-soliton profile by $6(u_0^2)_{xx}\big|_{x=0} = -\frac{3}{2}k^6 \neq 0$ **(c)** (`audit_onebody.py`). The test write-up `phase2/onebody_dispersion_test.tex` Eq. (5) correctly uses (4).

2. Inherited tadpole sign error in `phase2/oneloop.tex` (c). In the good equation (1), $\mathcal{H}_g = \frac{1}{2}\pi^2 + \frac{1}{2}u^2 - u^3 + \frac{1}{2}u_x^2$ contains $-u^3$. Wick normal ordering $:u^3: = u^3 - 3\langle u^2\rangle_0 u$ replaces $-u^3$ by $-:u^3: -3\langle u^2\rangle_0 u$, so the counterterm added to the bare energy is $+3\langle u^2\rangle_0 M$ **(c)**. In the bad equation (4), $\mathcal{H}_b$ contains $+u^3 = +:u^3: +3\langle u^2\rangle_0 u$, so passing to the normal-ordered Hamiltonian $:\mathcal{H}_b:$ adds the counterterm **(c)**

$$\Delta H_{\text{ct}}^{(b)} = -3\langle u^2\rangle_0^{\text{band}} \int_{-\infty}^{\infty} u_0(x)\,dx = -3\left(\frac{\hbar_B}{2\pi}\int_0^1 \frac{q\,dq}{\sqrt{1-q^2}}\right)(2k) = -\frac{3\hbar_B k}{\pi}, \tag{5}$$

where $\langle u^2\rangle_0^{\text{band}} = \frac{\hbar_B}{2\pi}$ on the stable band $|q| < 1$. In `phase2/oneloop.tex` Eq. (28), the good-equation sign $+3\langle u^2\rangle_0 M$ was carried over without flipping the cubic sign, producing $+\frac{3\hbar_B k}{\pi}$ (off by $+\frac{6\hbar_B k}{\pi}$). `phase2/onebody_dispersion_test.tex` §2.2 diagnoses and corrects this sign error **(c)** (`audit_onebody.py`).

2.2 Formal analytic continuation $F_{\text{ac}}(\beta)$ and light-root cancellation

3. Algebraic continuation from the good equation (c)/(d). For $\sigma = c^2 \in \{\pm 1\}$, the rescaled variables $\tilde{x} = cx$, $\tilde{t} = c^2 t$, $\tilde{\phi}(\tilde{x}, \tilde{t}) = c^{-1}\phi(x,t)$ satisfy **(c)**

$$\mathcal{L}_\sigma = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \sigma\phi_x^3 - \tfrac{1}{2}\sigma\phi_{xx}^2 = c^4\left(\tfrac{1}{2}\tilde{\phi}_{\tilde{t}}^2 - \tfrac{1}{2}\tilde{\phi}_{\tilde{x}}^2 + \tilde{\phi}_{\tilde{x}}^3 - \tfrac{1}{2}\tilde{\phi}_{\tilde{x}\tilde{x}}^2\right). \tag{6}$$

At $c = i$ ($\sigma = -1$), $\mathcal{L}_{-1} = \mathcal{L}_b$ and the good soliton parameter α maps to $i\beta$ ($k_g = \sin\alpha = i\sinh\beta = ik_b$, $v_g = \cos\alpha = \cosh\beta = v_b$, $P_{0,g}(i\beta) = iP_{0,b}(\beta)$). In `phase2/det_projected_oneloop.tex`, the one-loop projected determinant on the good branch gives **(c)**

$$F_g(k_g) := E_{1,g}\big|_P = \hbar_B\big(e(\alpha) - \cos\alpha\, p(\alpha)\big) = \hbar_B\left[\frac{\sqrt{3}}{\pi}\left(\frac{3}{4} - k_g^2\right)\operatorname{artanh}\frac{2k_g}{\sqrt{3}} - \frac{3k_g}{2\pi}\right]. \tag{7}$$

Applying the formal continuation prescription $F_{\text{ac}}(\beta) := -iF_g(ik_b)$ (**(d)** as a physical prescription for the ill-posed bad equation; **(c)** as an algebraic identity) yields **(c)**

$$\frac{F_{\text{ac}}(\beta)}{\hbar_B} = \frac{\sqrt{3}}{\pi}\left(k^2 + \frac{3}{4}\right)\arctan\frac{2k}{\sqrt{3}} - \frac{3k}{2\pi} = \frac{4}{3\pi}\beta^3 + \frac{14}{45\pi}\beta^5 + O(\beta^7), \qquad k = \sinh\beta. \tag{8}$$

Dividing by $|P_0(\beta)| = \frac{2}{3}\beta^3 + \frac{2}{3}\beta^5 + O(\beta^7)$ gives the small-amplitude limit **(c)** (`audit_onebody.py`)

$$C_{\text{ac}} := \lim_{\beta\to 0^+}\frac{F_{\text{ac}}(\beta)}{\hbar_B|P_0(\beta)|} = +\frac{2}{\pi} = 0.636619772367581343\ldots \tag{9}$$

4. Exact cancellation of the light (KdV) root in $F_g(\alpha)$ (c). A structural audit of (7) reveals *which modes* produce $C_{\rm ac} = +2/\pi$. In `phase2/oneloop_phase.tex` and `phase2/det_projected_oneloop.tex`, the phase-shift integrals $e(\alpha)$ and $p(\alpha)$ sum over the three scattering roots $\phi_1 = \alpha + \frac{2\pi}{3}$, $\phi_2 = \alpha - \frac{2\pi}{3}$ (heavy dispersive roots at wavenumber $|q| \sim \frac{\sqrt{3}}{2}$) and $\phi_3 = \alpha$ (light KdV root at $|q| \sim 0$). In the combination $e(\alpha) - \cos\alpha\, p(\alpha)$, the coefficient of $\frac{c_r}{4\pi}\log|\tan(\phi_r/2)|$ for root r is proportional to **(c)**

$$w(\phi_r) := 2\cos\alpha\sin\phi_r - \sin(2\phi_r) = 2\sin\phi_r(\cos\alpha - \cos\phi_r). \tag{10}$$

For the light root $\phi_3 = \alpha$, $w(\alpha) \equiv 0$ *identically* for all α **(c)**! For the heavy roots $\phi_{1,2} = \alpha \pm \frac{2\pi}{3}$, we have $w(\phi_1) = -w(\phi_2) = 2\sqrt{3}(\frac{3}{4} - \sin^2\alpha)$ **(c)** (`audit_onebody.py`). After continuation to $k = \sinh\beta$, the heavy-root arctangent term and the tadpole/endpoint term $-3k/(2\pi)$ expand as **(c)**

$$\frac{\sqrt{3}}{\pi}\left(\sinh^2\beta + \frac{3}{4}\right)\arctan\frac{2\sinh\beta}{\sqrt{3}} = \frac{3}{2\pi}\beta + \left(\frac{4}{3\pi} + \frac{1}{4\pi}\right)\beta^3 + O(\beta^5), \tag{11}$$

$$-\frac{3\sinh\beta}{2\pi} = -\frac{3}{2\pi}\beta - \frac{1}{4\pi}\beta^3 + O(\beta^5). \tag{12}$$

Thus the $O(\beta)$ terms cancel against the continuum Wick tadpole, and the surviving $+\frac{4}{3\pi}\beta^3$ term comes *100% from the short-wave dispersive roots* $|q| \sim \frac{\sqrt{3}}{2}$ *and the continuum UV tadpole*, with zero contribution from the long-wave KdV root ϕ_3 **(c)**.

2.3 Real stable-band trace $F_{\rm band}(\beta)$ and the differential annihilator $\mathcal{I}$

5. Stable-band quadrature and resonant jump (c). Integrating the exact scattering phase shift over the real stable band $|q| < 1$ gives **(c)**

$$\frac{F_{\rm band}(\beta)}{\hbar_B} = \frac{F_{\rm ac}(\beta)}{\hbar_B} + R(\beta), \qquad R(\beta) = -\frac{\sqrt{3}}{2}\left(\sinh^2\beta + \frac{3}{4}\right), \tag{13}$$

which we verified against 65-digit numerical quadrature of the transmission phase derivative $\partial_t \arg A(\sin t)$ at $\beta = 0.25$ and $\beta = 0.50$ to $< 10^{-52}$ in `audit_onebody.py` **(c)**. As noted in `onebody_dispersion_test.tex` §2.4, $R(\beta)$ originates from the co-moving resonance $\omega_b(q_*) = -vq_*$ at $q_* = -\frac{\sqrt{3}}{2}\,\mathrm{sgn}(v) \in (-1, 1)$, where the bad-equation group velocity $\omega_b'(q_*) = \frac{1-2q_*^2}{\sqrt{1-q_*^2}} = -\frac{1}{2}$ is opposite to v, flipping $\mathrm{sgn}(\omega_b'(q) + v)$ across q_* **(c)**. Because $R(0) = -\frac{3\sqrt{3}}{8} \neq 0$, the direct ratio $F_{\rm band}(\beta)/(\hbar_B|P_0(\beta)|) \sim -\frac{9\sqrt{3}}{16\beta^3}$ diverges as $\beta \to 0^+$ **(c)**. Applying $\mathcal{A}_\beta = \partial_\beta^3 - \partial_\beta$ gives $\mathcal{A}_\beta R(\beta) = -3\sqrt{3}\sinh\beta\cosh\beta$, which vanishes at $\beta = 0$ solely because $R(\beta)$ is an even function of β **(c)**. Hence $\mathcal{I}[F_{\rm band}] = \lim_{\beta\to 0^+}\frac{\mathcal{A}_\beta F_{\rm band}}{\hbar_B \mathcal{A}_\beta|P_0|} = +\frac{2}{\pi}$ isolates the odd part $F_{\rm ac}(\beta)$ algebraically, but does not remove the physical $O(1)$ threshold offset $R(0) = -\frac{3\sqrt{3}}{8}$ of the real stable-band integral **(c)**.

3 Audit of the quantum Toda Fermi-edge extraction

3.1 Exact non-perturbative finite-sea identities and analytical proof of $C_T = \frac{1}{2\pi}$

We now prove that the quantum Toda Fermi-edge quantity extracted numerically in `onebody_toda.py` is *analytically equal to* $\frac{1}{2\pi}$ **(c)**.

In the conventions of `phase2/frame_dictionary.tex` and `phase2/scripts/frame_edge.py`, for Planck constant $h = \hbar_T$, $t = 2\pi h$, Fermi point $q > 0$, kernel $K(\Delta) = 2\,\mathrm{Re}\,\psi(1 + i\Delta/h) + 2\ln h$, and phase $\Theta(\Delta) = \int_0^\Delta K(u)\,du$, the finite-sea Bethe operator on $[-q, q]$ is $(\mathcal{A}f)(x) = tf(x) - \int_{-q}^{q} K(x - z)f(z)\,dz$, with $\mathcal{A}\rho = 1$, $\mathcal{A}\eta = x$, density $n = 1/a = \int_{-q}^{q}\rho(x)\,dx$, dressed momentum derivative $D(p) = P'(p) = 1 + \int_{-q}^{q} K(p - x)\rho(x)\,dx$, dressed energy derivative $V(p) = E'(p) = p + \int_{-q}^{q} K(p - x)\eta(x)\,dx$, and sound speed $v_s(a, h) = V(q)/D(q) = \eta(q)/\rho(q)$ (since $D(q) = t\rho(q)$ and $V(q) = t\eta(q)$).

Proposition 1 (Exact endpoint identity (c)). *For every $q > 0$ and $h > 0$, the endpoint values $\rho(q)$ and $\eta(q)$ satisfy*

$$2at\,\rho(q)\eta(q) = 1 \qquad \Longleftrightarrow \qquad D(q)V(q) = \frac{\pi h}{a}. \tag{14}$$

Proof. Differentiating $(\mathcal{A}\rho)(x) = 1$ on $(-q, q)$ gives $(\mathcal{A}\rho')(x) = -\rho(q)\big(K(x-q) - K(x+q)\big)$ because $\rho(-q) = \rho(q)$. Since $\mathcal{A}$ is symmetric on $L^2[-q, q]$ and $\mathcal{A}\eta = x$, integration by parts yields

$$2q\rho(q) - \frac{1}{a} = \int_{-q}^{q} x\rho'(x)\,dx = \langle \mathcal{A}\eta, \rho' \rangle = \langle \eta, \mathcal{A}\rho' \rangle = -\rho(q) \int_{-q}^{q} \eta(x)\big(K(x-q) - K(x+q)\big)\,dx. \tag{15}$$

Using $(\mathcal{A}\eta)(\pm q) = \pm q \iff \int_{-q}^{q} K(x \mp q)\eta(x)\,dx = \pm\big(t\eta(q) - q\big)$ evaluates the right-hand side to $2q\rho(q) - 2t\rho(q)\eta(q)$. Cancelling $2q\rho(q)$ leaves $2t\rho(q)\eta(q) = \frac{1}{a}$ **(c)**. □

Identity (14) holds to 1.6×10^{-16} on a 64-node Gauss–Legendre solver at $(h, q) = (0.4, 1.5)$ (where the nearest digamma pole at $\pm ih$ lies at $\pm\frac{4}{15}qi$), to 4.3×10^{-31} on a 48-node solver at $(h, q) = (0.8, 1.0)$, and to $< 2 \times 10^{-12}$ across all saved rows of `frame_edge_data.json` (limited by `xtol=1e-13` on q in `frame_edge.py` line 27; `audit_onebody.py`) **(c)**. In addition, substituting (14) into $t\eta(q)[F_p(q) + F_p(-q)] = P(p) - V(p)$ simplifies the frame charge $\mathfrak{Q}_p$ of `frame_dictionary.tex` Eq. (14) to $\mathfrak{Q}_p = \overline{P}(p) - \big(V(p) - V(q)\big)$ and $V(p) - v_s D(p) = \overline{P}(p) - \frac{f_p^-}{a\rho(q)}$ **(c)**.

Proposition 2 (Exact scaling and Maxwell identities (c)). *For every $a, h > 0$ and $p > q(a, h)$,*

$$v_s(a, h) = -a\,\frac{\partial q(a, h)}{\partial a}\bigg|_h = \frac{a}{2}\left(q(a, h) - h\,\frac{\partial q(a, h)}{\partial h}\bigg|_a\right), \tag{16}$$

$$V(p, a, h) = \frac{\partial E(p, a, h)}{\partial p}\bigg|_{a,h} = \frac{\partial\big(aP(p, a, h)\big)}{\partial a}\bigg|_{p,h}, \tag{17}$$

$$\frac{V(p)}{D(p)} = \frac{\partial \overline{E}}{\partial \overline{P}}\bigg|_{a,h} = \frac{a}{2}\left(p - h\,\frac{\partial p}{\partial h}\bigg|_{\overline{P},a} - \frac{\overline{P}(p)}{D(p)}\right). \tag{18}$$

Proof. Differentiating $n(q, h) = \int_{-q}^{q} \rho(x)\,dx$ with respect to q at fixed h gives $\partial_q(1/a)|_h = 2t\rho(q)^2$. Combining with (14) gives $\partial_q a|_h = -2a^2 t\rho(q)^2 = -a/v_s$, proving the first equality in (16). Similarly, $\partial_q P(p)|_h = t\rho(q)\big(F_p(q) + F_p(-q)\big) = \frac{1}{2a\eta(q)}\big(P(p) - V(p)\big)$, which combines with $\partial_q a|_h = -a/v_s$ to yield (17). Finally, writing $K(\Delta) = K_0(\Delta/h) + 2\ln h$ with $K_0(u) = 2\,\mathrm{Re}\,\psi(1 + iu)$ and setting $x = h\xi$, $Q = q/h$ shows that $\tilde{\rho}(\xi) = t\rho(h\xi)$ solves $2\pi\tilde{\rho}(\xi) - \int_{-Q}^{Q} K_0(\xi - \zeta)\tilde{\rho}(\zeta)\,d\zeta = 1 + \frac{2\ln h}{a}$, so $q(a, h)/h = \mathcal{Q}(a + 2\ln h)$ and $aP(p, a, h) = h\hat{\mathcal{P}}(p/h, a + 2\ln h)$ depend on (a, h) only through the combination $s = a + 2\ln h$ **(c)**. Therefore $h\partial_h(q/h)|_a = 2\partial_a(q/h)|_h$, which gives the second equality in (16), and $h\partial_h(aP/h)|_{p/h,a} = 2\partial_a(aP/h)|_{p/h,h} = 2V(p)/h$, which rearranges at fixed $\overline{P} = P(p) - \pi h/a$ into (18) **(c)**. □

All three identities in Proposition 2 are verified to $< 10^{-25}$ at $(h, q) = (0.8, 1.0)$ ($N = 48$) in `audit_onebody.py` **(c)**.

Theorem 3 (Exact analytical values of $s_{1,T}$, $C_v(y)$, $C_E(y)$, and $C_T = \frac{1}{2\pi}$ (c)). *In the semiclassical limit $h \to 0$ at fixed $a > 0$ (with $b = \pi/a$):*

1. *Substituting the Fermi-edge expansion $q(a, h) = 2b - \frac{h}{2\pi}\ln\frac{h}{b} + c_q h + O(h^2 \ln h)$ into (16) cancels c_q and $\ln(h/b)$ identically:*

$$v_s(a, h) = \frac{a}{2}\left(2b + \frac{h}{2\pi} + O(h^2 \ln h)\right) = ab + \frac{ah}{4\pi} + O(h^2 \ln h) \implies s_{1,T} := \frac{3}{a}\lim_{h\to 0}\frac{v_s(a, h) - ab}{h} = \frac{3}{4\pi}. \tag{19}$$

2. *In the Fermi-edge scaling regime* $p = q(a,h) + hu(y)$ *where* $y = b\kappa^2/h$ *and* $\overline{P} = \frac{4b}{3a}\kappa^3(1 + O(\kappa^2)) = \frac{4}{3a}b^{-1/2}(yh)^{3/2}(1+O(yh/b))$, *equation* (18) *gives* $\lim_{h\to 0}\frac{V(p)/D(p)-v_s(a,h)}{h} = \frac{a}{6}y\big(u'(y)-1\big)$, *so in the inherited Boussinesq units of* `onebody_toda.py`*:*

$$C_v(y) = \frac{3}{4\pi} + \frac{1}{2}y\big(u'(y)-1\big), \qquad C_E(y) = \frac{3}{4\pi} + \frac{3}{4}y^{-3/2}\int_0^y \eta^{3/2}\big(u'(\eta)-1\big)\,d\eta. \tag{20}$$

3. *Using the Wiener–Hopf expansion (*`toda_wh_and_kp_synthesis.tex`*)* $u(y) = y - \frac{\ln y}{2\pi} + c_{\text{edge}} + \frac{d_0}{y} + \frac{e_0}{y^2} + O(y^{-3})$ *with* $d_0 = \frac{1}{48} - \frac{1}{4\pi^2}$ *and* $e_0 = \frac{4+6\zeta(3)-\pi^2}{96\pi^3}$ *yields*

$$C_v(y) = \frac{3}{4\pi} - \frac{1}{4\pi} - \frac{d_0}{2y} - \frac{e_0}{y^2} + O(y^{-3}) \xrightarrow{y\to\infty} \frac{1}{2\pi}, \tag{21}$$

$$C_E(y) = \frac{1}{2\pi} - \frac{3d_0}{2y} + B_0 y^{-3/2} + \frac{3e_0}{2y^2} + O(y^{-5/2}) \xrightarrow{y\to\infty} \frac{1}{2\pi}. \tag{22}$$

4. *In the opposite order of limits (*$h \to 0$ *first at fixed unscaled* $\overline{P} > 0$*, then* $\overline{P} \to 0^+$*),* $p(\overline{P}, h) = q(a,h) + b\kappa^2 + \frac{h}{2\pi}\ln\frac{h}{b\kappa^2} + c_{\text{edge}}h + o(h) = 2b + b\kappa^2 - \frac{h}{\pi}\ln\kappa + (c_q + c_{\text{edge}})h + o(h)$ *has no* $h\ln h$ *term, and* (18) *gives* $\lim_{\overline{P}\to 0^+}\lim_{h\to 0}\frac{3}{a}\frac{\overline{E}-E_{T,0}}{h\overline{P}} = \frac{1}{2\pi}$. *Hence the fixed-y and fixed-P iterated limits commute and both equal* $C_T = \frac{1}{2\pi} = 0.15915494309189535\ldots$ **(c)**.

3.2 Audit of `onebody_toda.py`: non-circularity and error-bar honesty

1. **Non-circularity (c):** Inspection of `phase2/scripts/onebody_toda.py`, `frame_edge.py`, and `frame_analysis.py` confirms that neither $\frac{1}{2\pi}$ nor $\frac{2}{\pi}$ enters any solver or fit.

2. **Numerical agreement with Theorem 3 (c):** The extrapolated sound-speed shift in `onebody_toda_results.json` is $s_{1,T}^{\text{num}} = 0.23873241714$ (quadratic fit) and 0.23873241473 (cubic fit), matching $\frac{3}{4\pi} = 0.23873241464$ to 2.5×10^{-9} and 9.6×10^{-11}, respectively. Across all nine grid points $y \in [0.1, 64]$, the independently extrapolated $C_v(y)$ in `onebody_toda_results.json` and $\frac{3}{4\pi} + \frac{1}{2}y\big(u'(y)-1\big)$ from `frame_analysis.json` agree to $< 2.8\times10^{-6}$ (`audit_onebody.py`).

3. **Honest error bars (c):** The central two-point $1/y$ extrapolations from $y \in \{32, 64\}$ give $C_{T,\infty}^{(E)} = 0.15911988$ and $C_{T,\infty}^{(v)} = 0.15906273$, which lie below $\frac{1}{2\pi} = 0.15915494$ by -3.51×10^{-5} and -9.22×10^{-5}. Both offsets are well inside the central edge-propagated error bars ($\pm1.84\times10^{-4}$ and $\pm5.26\times10^{-4}$) and roughly 100× smaller than the conservative 12-fit envelopes ($\pm3.34\times10^{-3}$ and $\pm9.97\times10^{-3}$). The slight downward offset is explained by higher-order $y^{-3/2}$ and y^{-2} terms in (21)–(22) together with residual finite-κ curvature at $y = 64$, $h = 0.02$ ($\kappa = \sqrt{yh/b} \approx 1.45$).

4 Systematic hunt for a factor of 4 in the Toda–Boussinesq dictionary

Having established that $C_{\text{ac}} = \frac{2}{\pi}$ and $C_T = \frac{1}{2\pi}$ are both exact within their respective definitions, so that $C_{\text{ac}}/C_T = 4$ *exactly* **(c)**, we trace every normalisation in the Toda–bad-Boussinesq dictionary (`phase2/toda_route.tex`, `phase2/scripts/toda_scaling_fast.py`, `phase2/audit_toda_dictionary.tex`) to test whether any convention introduces a hidden factor of 4.

Detailed audit of the six candidates (c):

1. **Action and Planck constant ($\hbar_T = 3b\hbar_B$):** Under $x_B = \lambda_x n$ ($\lambda_x = 2\sqrt{3}$), $t_B = \lambda_t t_T$ ($\lambda_t = 2\sqrt{3}b$), and $u_n = \lambda_\phi \phi_B$ ($\lambda_\phi = -\sqrt{3}$), the continuum expansion of the Toda Lagrangian $L_T = \sum_n[\frac{1}{2}\dot{u}_n^2 - b^2(e^{u_n-u_{n-1}} - 1 - (u_n - u_{n-1}))]$ is $\mathcal{L}_T = \frac{1}{2}u_{t_T}^2 - \frac{b^2}{2}u_n^2 + \frac{b^2}{6}u_n^3 + \frac{b^2}{24}u_{nn}^2 + O(\partial^6)$.

Dictionary candidate	Exact relation between Toda (T) and bad Boussinesq (B)	Factor on C_T	Status
1. $\hbar_T = 3b\hbar_B$ & lattice scaling	$x_B = 2\sqrt{3}\,n$, $t_B = 2\sqrt{3}b\,t_T$, $u_n = -\sqrt{3}\phi_B \Rightarrow S_T = 3b\,S_B$, $\hbar_T = 3b\hbar_B$	1	Verified **(c)**
2. Symplectic 2-form Ω	$\Omega_T = \sum_n dp_n \wedge du_n = 3b \int d\pi_B \wedge d\phi_B = 3b\,\Omega_B \Rightarrow \Omega_T/\hbar_T = \Omega_B/\hbar_B$	1	Verified **(c)**
3. Momentum P definitions	$P_n = a\overline{P} = 6\sqrt{3}b\,P_B(1+O(\beta^2))$; Fermi-edge subtraction $\overline{P}(p) = P_{\rm dr}(p) - \frac{\pi\hbar_T}{a}$ vanishes at $p=q$	1	Verified **(c)**
4. Energy E_1 & chemical potential	$\overline{E}(p) = E_{\rm dr}(p) - E_{\rm dr}(q) = 6\sqrt{3}b^2E_B(1+O(\beta^2))$; combined ratio $\frac{E_B - E_{B,0}}{\hbar_B P_B} = \frac{3}{a}\frac{\overline{E}-E_{T,0}}{\hbar_T\overline{P}}$	1	Verified **(c)**
5. Mass $M = 2k$ & soliton width	$M_T = -2\kappa = -\sqrt{3}M_B$ ($\kappa = \sqrt{3}\beta$ in Phase 2; Phase 1 $\kappa_{\rm P1} = 2\kappa$ cancels in $E_{T,0}(P_{T,0})$)	1	Verified **(c)**
6. Factor 3 in $\delta_T \to 3b\,\delta_{\rm bad}$	$\delta_T = 3b\,\delta_{\rm bad}(1+O(\beta^2)) \Rightarrow \delta_T/\hbar_T = \delta_{\rm bad}/\hbar_B$	1	Verified **(c)**

Table 1: Audit of all six candidate normalisations in the Toda–bad-Boussinesq dictionary. Every candidate introduces an exact conversion factor of 1 (`audit_onebody.py`).

Changing variables $(n, t_T) \mapsto (x_B, t_B)$ gives $S_T = \int dt_T\, dn\, \mathcal{L}_T = \frac{\lambda_\phi^2\lambda_t}{\lambda_x}\int dt_B\, dx_B\, \mathcal{L}_B = 3b\,S_B$. Hence $S_T/\hbar_T = S_B/\hbar_B$ requires $\hbar_T = 3b\hbar_B$, with factor 1 **(c)**.

2. **Symplectic normalisation:** With $p_n = \partial\mathcal{L}_T/\partial\dot{u}_n = \dot{u}_n = \lambda_\phi\lambda_t\pi_B$ and $dn = dx_B/\lambda_x$, $\Omega_T = \int dn\, dp_n \wedge du_n = \frac{\lambda_\phi^2\lambda_t}{\lambda_x}\int dx_B\, d\pi_B \wedge d\phi_B = 3b\,\Omega_B$. Thus $\Omega_T/\hbar_T = \Omega_B/\hbar_B$: factor 1 **(c)**.

3. **Momentum definitions (Π_T, P_n, $\overline{P}$, and $P_{\rm dr}(q) = \pi\hbar_T/a$):** As proved in `phase2/audit_toda_dictiona` the mechanical momentum $\Pi_T = \sum_n p_n$ and the Noether label momentum $P_n = -\int dn\, p_n u_n'$ are distinct conserved charges; the Bethe excitation momentum above the Fermi edge $\overline{P}(p) = P_{\rm dr}(p) - P_{\rm dr}(q)$ is related to the label momentum by $P_n = a\overline{P} = \frac{4}{3}b\kappa^3 + \frac{4}{15}b\kappa^5 + O(\kappa^7)$. Under $\kappa = \sqrt{3}\beta$, $P_n = 6\sqrt{3}b(\frac{2}{3}\beta^3 + \frac{2}{5}\beta^5 + O(\beta^7)) = 6\sqrt{3}b\,P_B(1 - \frac{2}{5}\beta^2 + O(\beta^4))$. At the small-amplitude limit $\beta \to 0$, $P_B = \frac{a}{6\sqrt{3}b}\overline{P}(1+O(\beta^2))$, and subtracting $P_{\rm dr}(q) = \frac{\pi\hbar_T}{a}$ sets $\overline{P}(q) = 0$ at the soliton threshold: factor 1 **(c)**.

4. **Energy E_1, chemical potential, and conversion factor $\frac{3}{a}$:** The excitation energy above the Fermi sea is $\overline{E}(p) = E_{\rm dr}(p) - E_{\rm dr}(q)$, where $\mu = q^2/2$ is already absorbed because adding a particle at fixed N or fixed μ shifts the threshold energy by $E_{\rm dr}(q) = 0$. Classically, $\overline{E}_{T,0}(\kappa) = 2b^2(\sinh\kappa\cosh\kappa - \kappa) = 6\sqrt{3}b^2(\frac{2}{3}\beta^3 + \frac{2}{5}\beta^5 + O(\beta^7)) = 6\sqrt{3}b^2E_{B,0}(1+O(\beta^2))$. Combining $E_B = \frac{\overline{E}}{6\sqrt{3}b^2}$, $P_B = \frac{a\overline{P}}{6\sqrt{3}b}$, and $\hbar_B = \frac{\hbar_T}{3b}$ yields

$$\frac{E_B - E_{B,0}(P_B)}{\hbar_B P_B} = \frac{\big(\overline{E} - E_{T,0}(\overline{P})\big)/(6\sqrt{3}b^2)}{(\hbar_T/3b)\cdot a\overline{P}/(6\sqrt{3}b)} = \frac{3}{a}\,\frac{\overline{E} - E_{T,0}(\overline{P})}{\hbar_T\overline{P}}, \tag{23}$$

matching Eq. (17) of `onebody_dispersion_test.tex` with conversion factor 1 **(c)**.

5. **Mass $M = 2k$ and soliton width conventions:** In Phase 1 (`phase1/scripts/toda_limit.py`), $\tau_n = 1 + e^{\kappa_{\rm P1}n - \omega t}$ uses $\kappa_{\rm P1} = \sqrt{12}\beta = 2\kappa$, whereas all Phase 2 scripts use $r_n - b^2 = b^2\sinh^2\kappa\,\mathrm{sech}^2(\kappa n - \dots)$ with $\kappa = \sqrt{3}\beta$. In Phase 2 conventions, $M_T = u_\infty - u_{-\infty} = -2\kappa = -2\sqrt{3}\beta = -\sqrt{3}M_B(1+O(\beta^2)) = \lambda_\phi M_B$. Moreover, κ is eliminated when expressing $\overline{E}_{T,0}(\overline{P}) = ab\overline{P} + \frac{a}{10}\left(\frac{3a}{4b}\right)^{2/3} b\overline{P}^{5/3} + \dots$, so κ conventions cannot alter $\lim_{\overline{P}\to 0}\frac{\overline{E}-E_{T,0}}{h\overline{P}}$: factor 1 **(c)**.

6. **The factor 3 in $\delta_T \to 3b\,\delta_{\rm bad}$:** At small amplitude, the head-on two-soliton phase shift is $\delta_{\rm bad} \sim 4\beta_1\beta_2$ and $\delta_T \sim 4b\kappa_1\kappa_2 = 12b\beta_1\beta_2 = 3b\,\delta_{\rm bad}$. Since $\hbar_T = 3b\hbar_B$, the quantum phase $\delta_T/\hbar_T = \delta_{\rm bad}/\hbar_B$ has factor 1 **(c)**.

Conclusion on the factor of 4: Verdict (b) (c). No missing factor of 4 exists anywhere in the Toda–Boussinesq dictionary. Instead, both numbers are exact in their own theories—$C_{\rm ac} = +\frac{2}{\pi}$ and $C_T = +\frac{1}{2\pi}$—and their ratio is 4 for a clear physical reason:

- On the Toda side, $C_T = \frac{1}{2\pi} = \frac{3}{4\pi} - \frac{1}{4\pi}$ is the sum of the lattice ground-state sound-speed renormalisation $s_{1,T} = +\frac{3}{4\pi}$ (from the finite-density Fermi sea $[-q, q]$) and the universal logarithmic Fermi-edge dressing shift $-\frac{1}{4\pi}$ from $u(y) = y - \frac{\ln y}{2\pi} + c_{\rm edge} + O(y^{-1})$. Subtracting the quantum sound speed gives $C_T - s_{1,T} = -\frac{1}{4\pi} < 0$.
- On the continuum side, the acoustic speed is unrenormalised ($s_{1,\rm cont} = 0$) and, by (10), the long-wave KdV root $\phi_3 = \alpha$ contributes *zero* to $F_g(\alpha) = e(\alpha) - \cos\alpha\, p(\alpha)$. The value $C_{\rm ac} = +\frac{2}{\pi}$ comes entirely from the short-wave dispersive roots $\phi_{1,2}$ at $|q| \sim \frac{\sqrt{3}}{2}$ (where the bad Boussinesq dispersion relation $\omega_b(q) = |q|\sqrt{1-q^2}$ turns over and enters the unstable band $|q| > 1$) combined with the continuum UV Wick tadpole. Relative to the sound-subtracted Toda slope $-\frac{1}{4\pi}$, the ratio is $C_{\rm ac}/(C_T - s_{1,T}) = -8$ **(c)**.

5 Verdict table and failures recorded

Audited claim / target	Label	Verdict	Independent audit finding
`report.tex` line 69 bad Lagrangian $+\phi_x^3$ sign	**(c)**	**Refuted**	Gives $u_{tt} = u_{xx} + u_{xxxx} - 3(u^2)_{xx}$; fails bad PDE on u_0 by $-\frac{3}{2}k^6$ at $x = 0$. Correct bad $\mathcal{L}_b$ has $-\phi_x^3$.
`oneloop.tex` Eq. (28) bad tadpole $+\frac{3\hbar k}{\pi}$	**(c)**	**Refuted**	$\mathcal{H}_b$ has $+u^3$, so Wick ordering gives $-\frac{3\hbar_B k}{\pi}$ (off by $\frac{6\hbar_B k}{\pi}$). Fix in `onebody_dispersion_test.tex` confirmed.
Formal continuation $F_{\rm ac}/(\hbar_B\lvert P_0\rvert) \to +2/\pi$	**(c)**/**(d)**	**Confirmed**	Algebra **(c)** verified; physical continuation to ill-posed bad branch is **(d)**. Comes 100% from heavy roots $\lvert q\rvert \sim \sqrt{3}/2$.
Real stable-band trace $F_{\rm band}$	**(c)**	**Confirmed**	$F_{\rm band} = F_{\rm ac} - \frac{\sqrt{3}}{2}(k^2 + \frac{3}{4})$; $F_{\rm band}/\lvert P_0\rvert \to -\infty$, while $\mathcal{I}[F_{\rm band}] = +2/\pi$ only because $R(\beta)$ is even.
Toda extraction non-circularity & error bars	**(c)**	**Confirmed**	Non-circular. Exact analytical limit proved to be $C_T = \frac{1}{2\pi} = 0.1591549\ldots$ (with $s_{1,T} = \frac{3}{4\pi}$), inside 0.15912 ± 0.00334.
Dictionary factor-of-4 hunt	**(c)**	**Verdict (b)**	All 6 dictionary candidates give factor 1. The ratio $C_{\rm ac}/C_T = 4$ (or -8 after sound subtraction) is physical/structural.

Table 2: Summary verdict table for the independent audit (`phase2/scripts/audit_onebody.py`).

Failures recorded.

1. **Bad-equation cubic sign typo in `phase2/report.tex` (line 69):** Confirmed failure in existing documentation ($+\phi_x^3$ printed instead of $-\phi_x^3$); encoded as a non-vanishing residual assertion in `audit_onebody.py`.
2. **Bad-equation Wick tadpole sign error in `phase2/oneloop.tex` Eq. (28):** Confirmed failure in earlier write-up ($+3\hbar_B k/\pi$ instead of $-3\hbar_B k/\pi$), already caught and recorded in `onebody_dispersion_test.tex` §2.2; encoded in `audit_onebody.py`.

3. **Heavy-root weight factor of 2 in initial draft of `audit_onebody.py`:** In the first run of `audit_onebody.py` (line 124), we asserted $w(\phi_1) - w(\phi_2) = 2\sqrt{3}(\frac{3}{4} - \sin^2\alpha)$ instead of $4\sqrt{3}(\frac{3}{4} - \sin^2\alpha)$ because $w(\phi_1) = -w(\phi_2) = 2\sqrt{3}(\frac{3}{4} - \sin^2\alpha)$ already carries the factor of 2 before taking $w(\phi_1) - w(\phi_2)$. Fixed in `audit_onebody.py`.

4. **Bernstein-ellipse quadrature tolerance at $(h, q) = (0.4, 1.5)$, $N = 64$ in `audit_onebody.py`:** In the second run of `audit_onebody.py`, the assertion `id1_err_64 < 1e-16` at $(h, q) = (0.4, 1.5)$ failed because the 64-node Gauss–Legendre truncation error for a kernel pole at $\pm\frac{4}{15}qi$ is 1.59×10^{-16} (just above 10^{-16}, consistent with `frame_identities.json` line 21), and the endpoint derivative $\partial_q a$ at $N = 48$, $(0.4, 1.5)$ has truncation error $\sim 3 \times 10^{-11}$. Relaxed the $(0.4, 1.5)$, $N = 64$ bound to 5×10^{-16}, moved the Proposition 2 finite-difference tests to $(h, q) = (0.8, 1.0)$ at $N = 48$, and corrected a missing factor of h on $\partial_h P|_{p,a}$ in line 304 of `audit_onebody.py`, after which Proposition 1 and Proposition 2 hold to $< 10^{-25}$.

5. **Transmission phase-derivative integrand in stable-band quadrature check:** An intermediate edit to `audit_onebody.py` used a KdV-like two-arctangent approximation for the bad-equation phase shift instead of the exact Boussinesq transmission phase derivative $\partial_t \arg A(\sin t)$ with $A = (Z + iX)/(Z - iX)$, $X = 3kq$, $Z = v\,\mathrm{sgn}(q)\sqrt{1 - q^2} - 1 - 2k^2 + 2q^2$. Replaced with the exact Boussinesq phase derivative, which matches $F_{\mathrm{band}}(\beta)$ to $< 10^{-52}$.

References

[1] E. K. Sklyanin, *The quantum Toda chain*, Lecture Notes in Physics **226** (1985), 196–233 (used via the verified Bethe equations in `phase2/toda_route.tex`; full text not in `literature/`).